

Outline

What is SDN?

OpenFlow basics

Why is SDN happening now? (a brief history)

4D discussion

What is SDN?

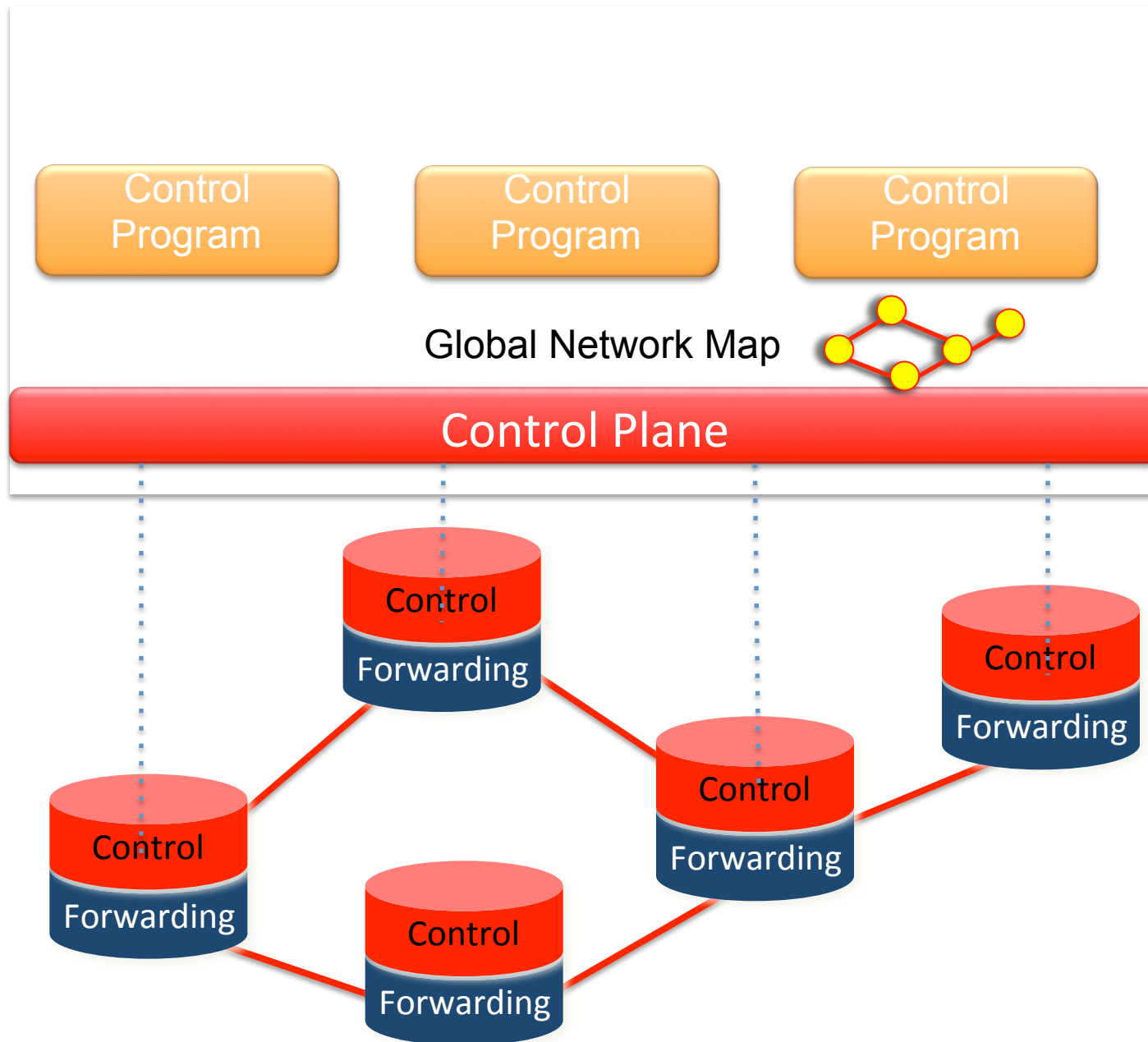
Software Defined Network

A network in which the control plane is physically separate from the data plane.

and

A single (logically centralized) control plane controls several forwarding devices.

Software Defined Network (SDN)



What You Said

“Overall, the idea of SDN feels a little bit unsettling to me because it is proposing to change one of the main reasons for the success of computer networks: fully decentralized control. Once we introduce a centralized entity to control the network we have to make sure that it doesn’t fail, which I think is very difficult.”

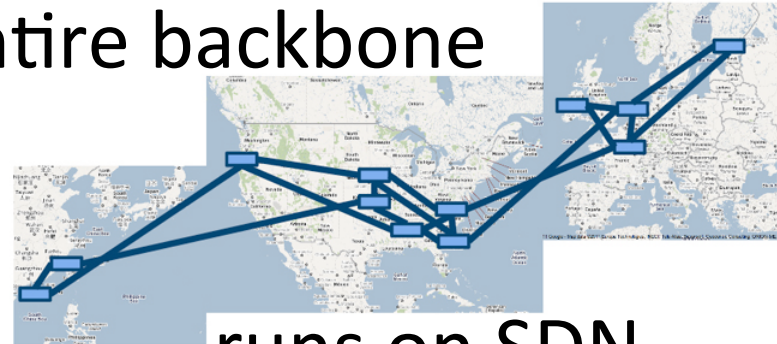
A Major Trend in Networking



OPEN NETWORKING
FOUNDATION



Entire backbone



runs on SDN

nicira

Bought for **\$1.2 billion**
(mostly cash)

The Networking “Planes”

Data plane: processing and delivery of packets with local forwarding state

- Forwarding state + packet header → forwarding decision
- Filtering, buffering, scheduling

Control plane: computing the forwarding state in routers

- Determines how and where packets are forwarded
- Routing, traffic engineering, failure detection/recovery, ...

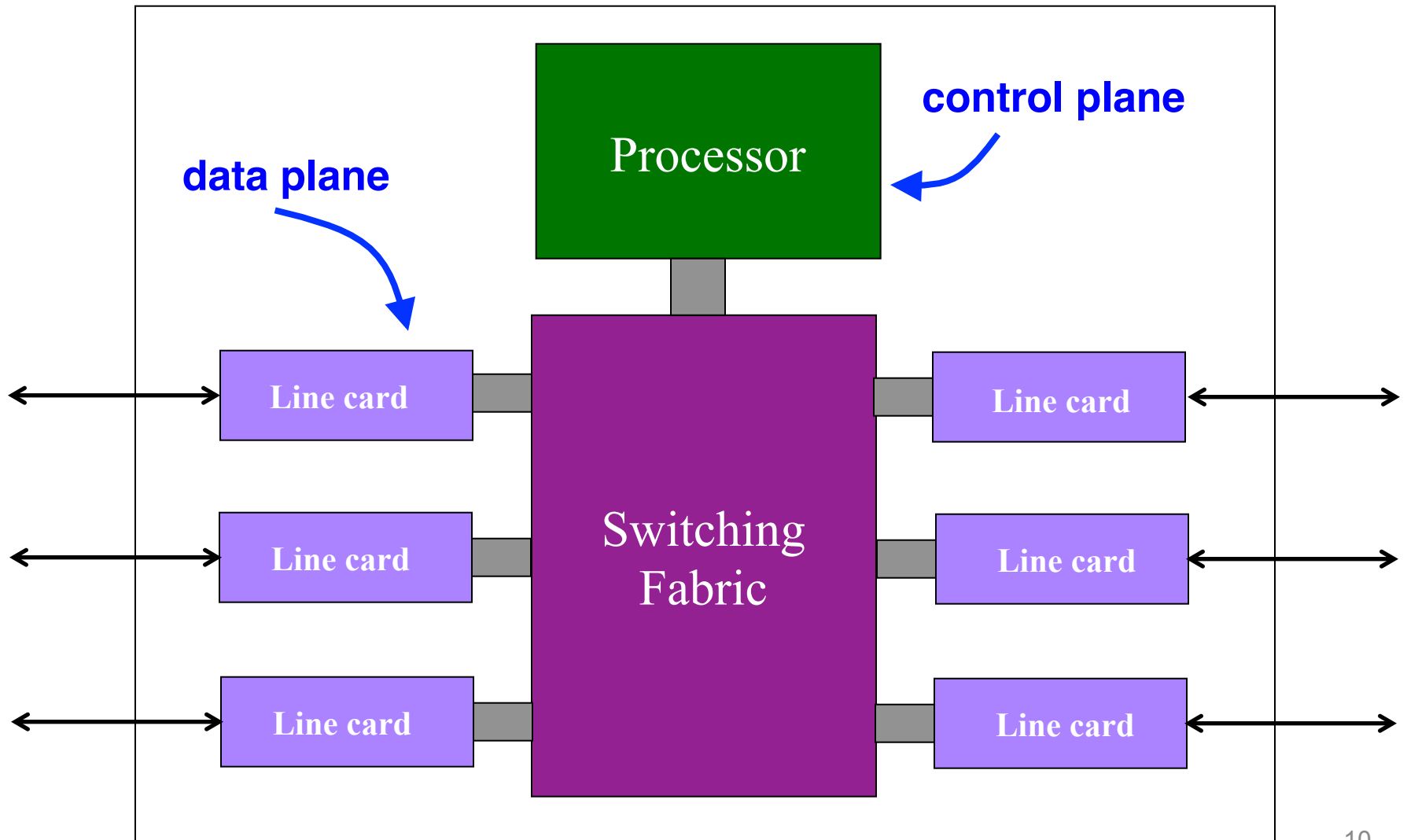
Management plane: configuring and tuning the network

- Traffic engineering, ACL config, device provisioning, ...

Timescales

	Data	Control	Management
Time-scale	Packet (nsec)	Event (10 msec to sec)	Human (min to hours)
Location	Linecard hardware	Router software	Humans or scripts

Data and Control Planes

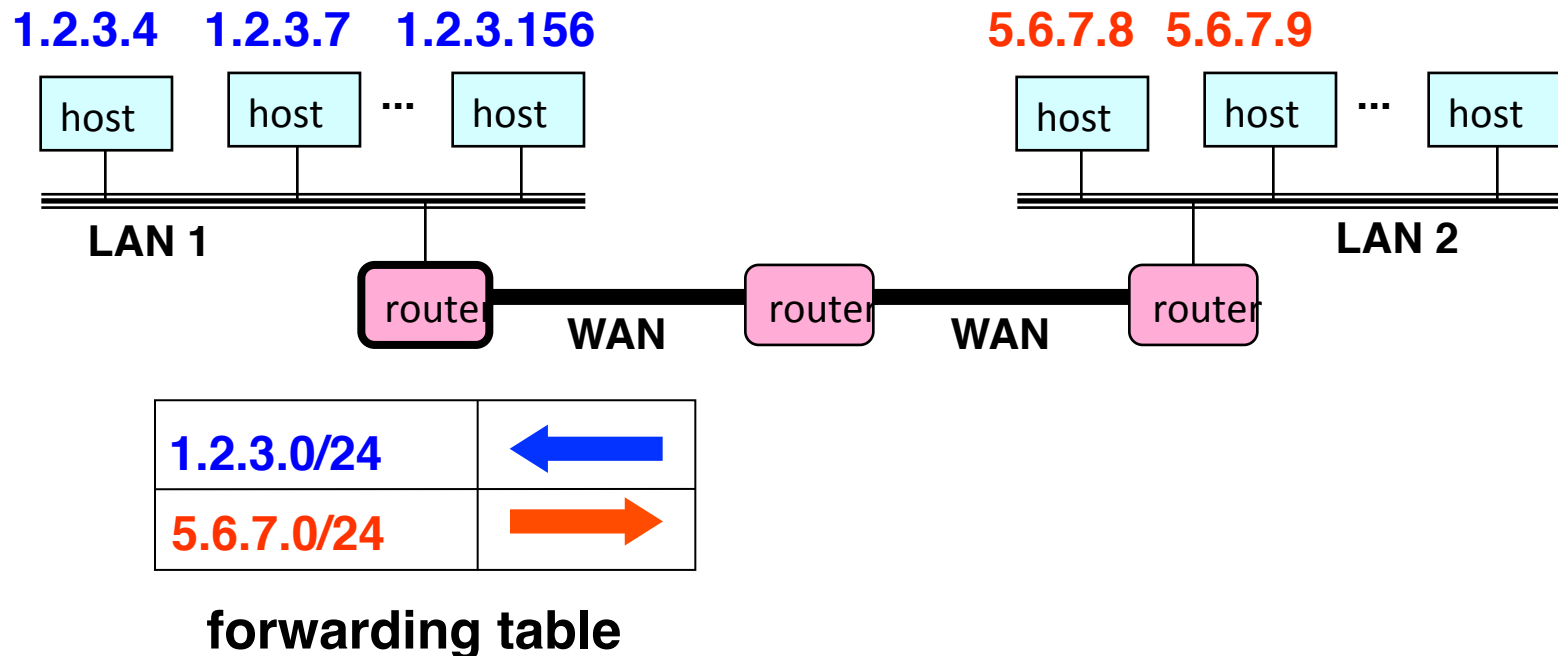


Data Plane

Streaming algorithms on packets

- Matching on some header bits
- Perform some actions

Example: IP Forwarding



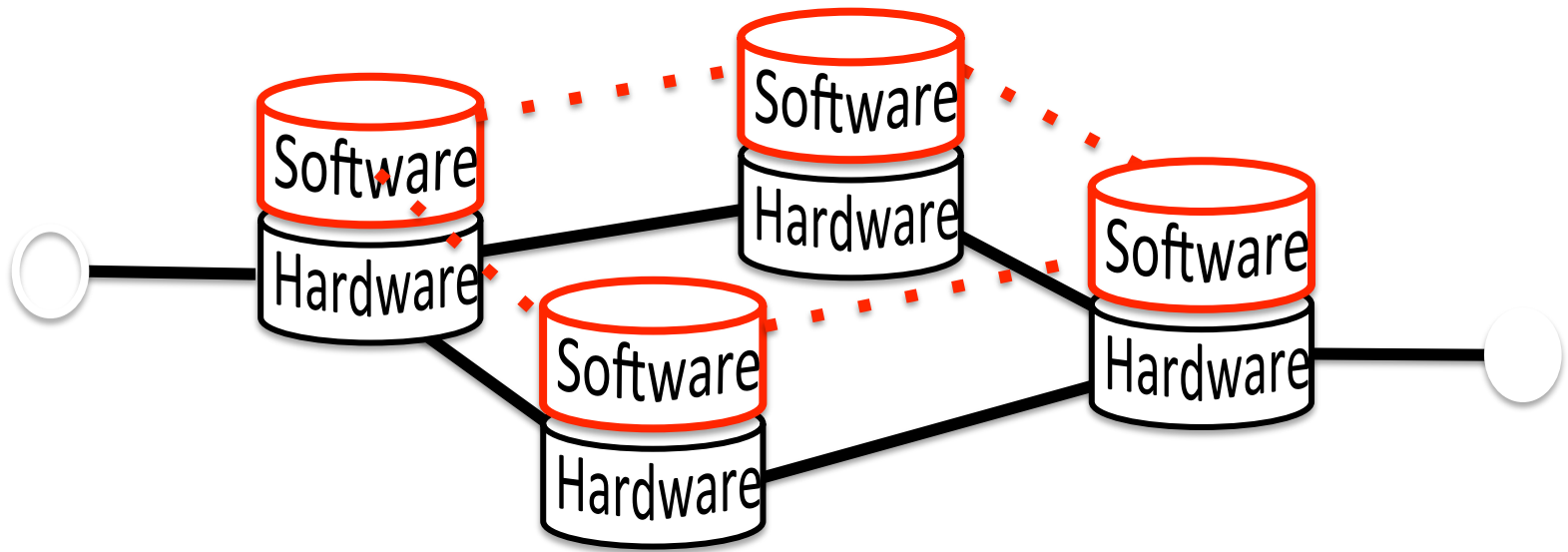
Control Plane

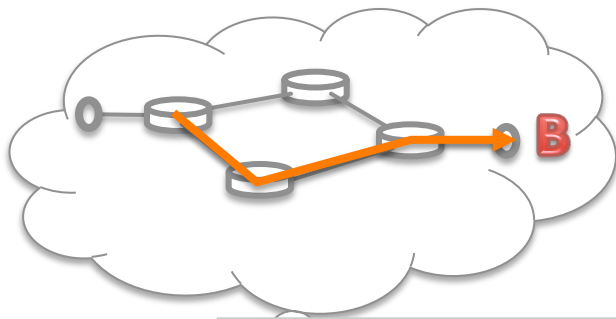
Compute paths the packets will follow

- Populate forwarding tables
- Traditionally, a distributed protocol

Example: **Link-state routing (OSPF, IS-IS)**

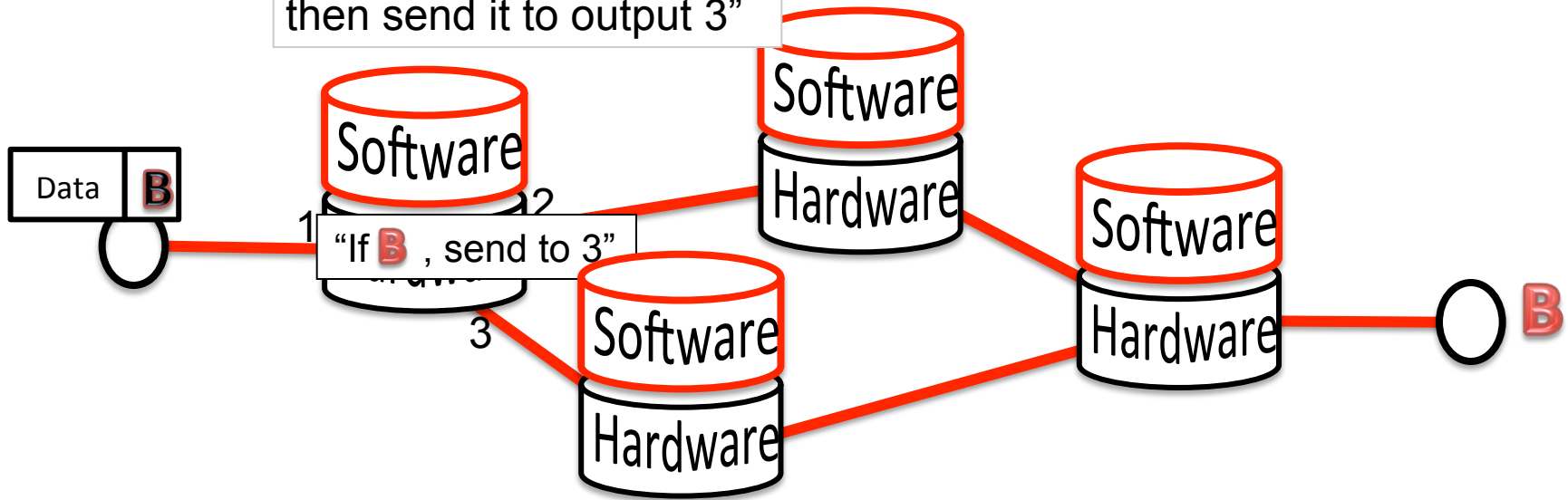
- Flood the entire topology to all nodes
- Each node computes shortest paths
- Dijkstra's algorithm





1. Figure out which routers and links are present.
2. Run Dijkstra's algorithm to find shortest paths.

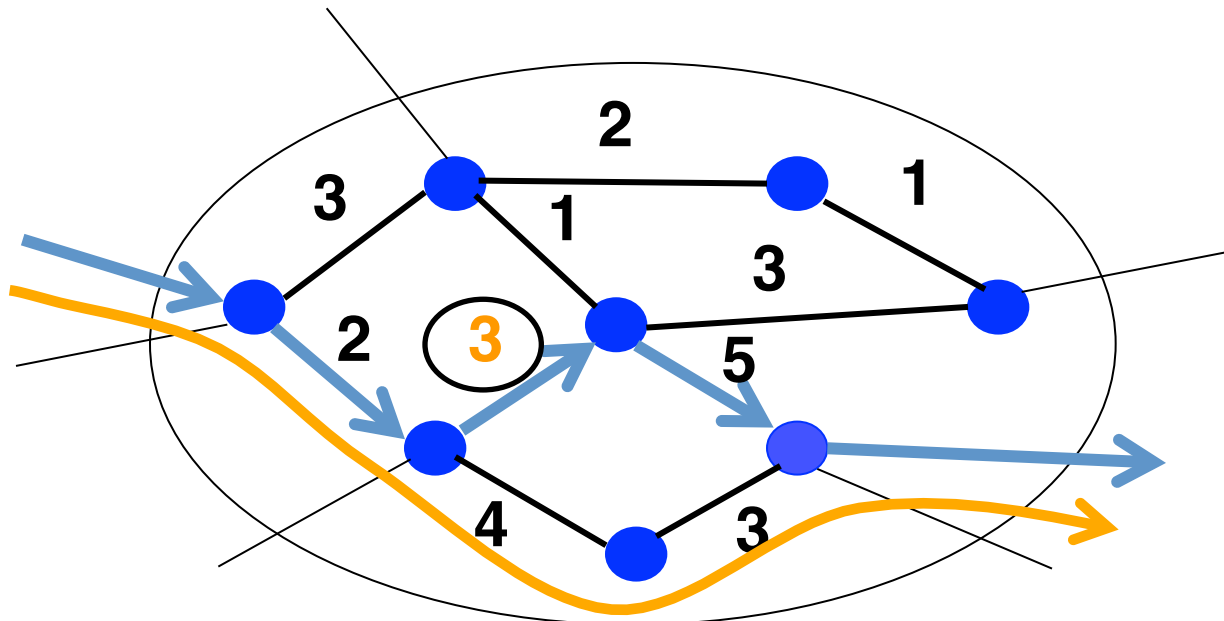
"If a packet is going to B,
then send it to output 3"



Management Plane

Traffic Engineering: setting the weights

- Inversely proportional to link capacity?
- Proportional to propagation delay?
- Network-wide optimization based on traffic?



Challenges

(Too) many task-specific control mechanisms

- No modularity, limited functionality

Indirect

- Mus
- Ex.

The network is

- Hard to reason about
- Hard to evolve
- Expensive

ou want

Uncoordinated

- Can

Interacting protocols and mechanisms

- Routing, addressing, access control, QoS

Example 1: Inter-domain Routing

Today's inter-domain routing protocol, BGP, artificially constrains routes

- Routing only on **destination IP address blocks**
- Can only influence **immediate neighbors**
- Very difficult to incorporate other information

Application-specific peering

- Route video traffic one way, and non-video another

Blocking denial-of-service traffic

- Dropping unwanted traffic further upstream

Inbound traffic engineering

- Splitting incoming traffic over multiple peering links